Educators Guide for *mPIXdaq*

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Data acquisition, visualization and analysis for the Advacam *miniPIX* (EDU) silicon pixel detector

Vers. 1.1, April 2026 DOI 10.5281/zenodo.19280859

This document is a guide for educators who want to explore the possibilitieslt3 of a modern radiation detection sensor and its implications on new ways to teach the subject of radioactivity at secondary and high-school level.

While installation, general purpose and use of the *mPIXdaq* software as well as the options for data acquisition, data analysis and visualization are described in the *README* document of the *mPIXdaq* package, this guide focuses on practical applications in laboratory courses.

The miniPIX (EDU) and its educational potential

The miniPIX (EDU) device is a modern silicon pixel detector that precisely measures the spacial distribution and magnitude of energy depositions caused by ionizing particles traversing the sensitive volume consisting of 256x256 pixels of 55x55x300µm³ volume each.
Segmented silicon detectors like the *miniPIX* were originally developed for tracking of charged particles in high-energy physics. They are now also widely used in diverse fields for particle detection, radiation dose measurements and imaging in material sciences and in medical applications.

The affordable miniPIX (EDU) variant from *Advacam* is based on the *Timepix* hybrid silicon pixel device developed at CERN by the Medipix collaboration. More details on the detector and its diverse applications may be found in the *Medipix CERN brochure*.

Thanks to the USB interface and the availability of a user program and drivers for various platforms, the detector is easy to use and allows data visualization in real-time. It is particularly useful to interactively investigate the properties and interactions with matter of α, β and γ radiation and of muons from cosmic rays in educational contexts, opening new and very intuitive ways of teaching nuclear and particle physics. The visual impression of recorded energy depositions in the pixel sensor resembles images produced by cloud chambers, however, the *miniPIX* is much easier to set-up and use and offers the additional advantage that quantitative, digital information with accurate spatial resolution of the deposited energy is also available. Besides visual inspection, recorded data sets can thus be analyzed in detail to quantitatively explore the properties of radiation.

The *miniPIX* is a cutting-edge technological masterpiece combining 65536 individual, radiation-sensitive pixels, arranged in an array of shape 256x256. The size of the sensitive volume of the pixel sensor is 14.1 x 14.1 x 0.3 mm³, segmented into 256 x 256 square pixels covering an area of 55 µm² each. The chip is covered by a very thin aluminum foil of 0.5µm thickness, and, in addition, there is a dead silicon layer thinner than 1µm in front of the sensitive volume. Bump-bonded to the sensor is a readout chip connecting each element of the pixel matrix to its own preamplifier, discriminator and digital counter integrated on the readout chip.

The deposited energy in each pixel is displayed as color-coded pixel in the form of a two-dimensional image. Such images of different types of radiation give a direct impression of the ways how radiation interacts with matter: very localized energy deposits for α particles, long, bent traces from β particles, and - typically small - energy deposits from γ rays, which also stem from electrons (i.e. β particles) produced via the Compton process or the photo effect. At rather low rates, also muons from cosmic rays are observed, which are characterized by straight tracks with constant mean ionization along the trace.

A schematic view of a charged-particle track in the sensitive detector material and the projection of the energy deposits on the pixel readout-plane is shown in the figure below.

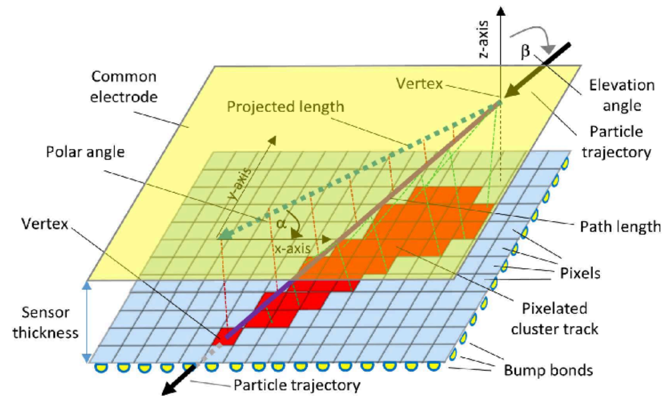


Abbildung 1: 3d-View of pixel cells and example of a particle track

The basic operating principle of the detector depends on the p-n junction of p- and n-type silicon semiconductors and should be well-known to students. It is analogous to other typical sensors for visible or infrared light, i.e. photo-diodes, solar cells or pixels in digital cameras. Electron-hole pairs are produced by charged particles traversing the charge-depleted zone of the p-n junction, and their number and hence the collected charge is proportional to the energy deposited by the traversing particle in the sensitive volume of the pixel.

Compared to other detection techniques, which simply count the occurrences of single particle interactions in a large volume, the *miniPIX* is special because it records with high spatial resolution all energy deposits that occurred within a freely chosen exposure time. During exposure, all energy deposits in each pixel summed up by measuring the time the pixel signal exceeds a given threshold. If the total number of objects hitting the detector is sufficiently small, the total energy deposit in each pixel is associated to one particle only.

A schematic of the detector is shown below.

The chip is read out in the so-called “frame mode”, i.e. all pixels are read out at the same time, providing one frame with deposited energies per pixel. If operated in time-over-threshold (“*ToT*”) mode, returned pixel readings represent the time the signal is over a given threshold in counts of the chip clock running at a frequency of approximately 10 MHz. *ToT* is linearly related to the energy deposition for large deposits exceeding 50 keV. The functional dependence on the deposited energy E, including threshold effects, is approximated by the following function

$$ToT \;=\; a\,E + b - c/(E - t)$$

Typical values of the calibrations constants are a = 1.6, b=23, c=23 and t=4.3. Each pixel has its individual calibration stored in a calibration file that is proviced together with each device and loaded at initialization time. This calibration is applied to obtain pixel readings in units of keV. The calibration is reliable up to pixel energies of about one MeV; beyond this the response shows a strong nonlinearity, overshooting between about one and two MeV and saturating beyond two MeV. For details, see the articles by J. Jakubek, *Precise energy calibration of pixel detector working in time-over-threshold mode*, NIM A 633 (2011), 5262-5265, and M. Sommer et al., *High-energy per-pixel calibration of timepix pixel detector with laboratory alpha source*, NIM A 1022 (2022) 165957.

Its high sensitivity and spatial mapping of energy deposits as well as the digitally recorded data make the *miniPIX* superior to other, classical detectors. While it allows the same typical text-book measurements to be made, like measurements of rates or penetration depths of different types of radiation, more extensive possibilities open up:

- direct observation of how radiation interacts with the silicon,
- discrimination of radiation types based on the pixel patterns,
- demonstration of Poisson statistics by counting objects in the recorded frames,
- dependence of the measured rates on the distance to the source,
- energy measurements of α particles and of their energy loss in matter,
- energy loss of β particles in matter,

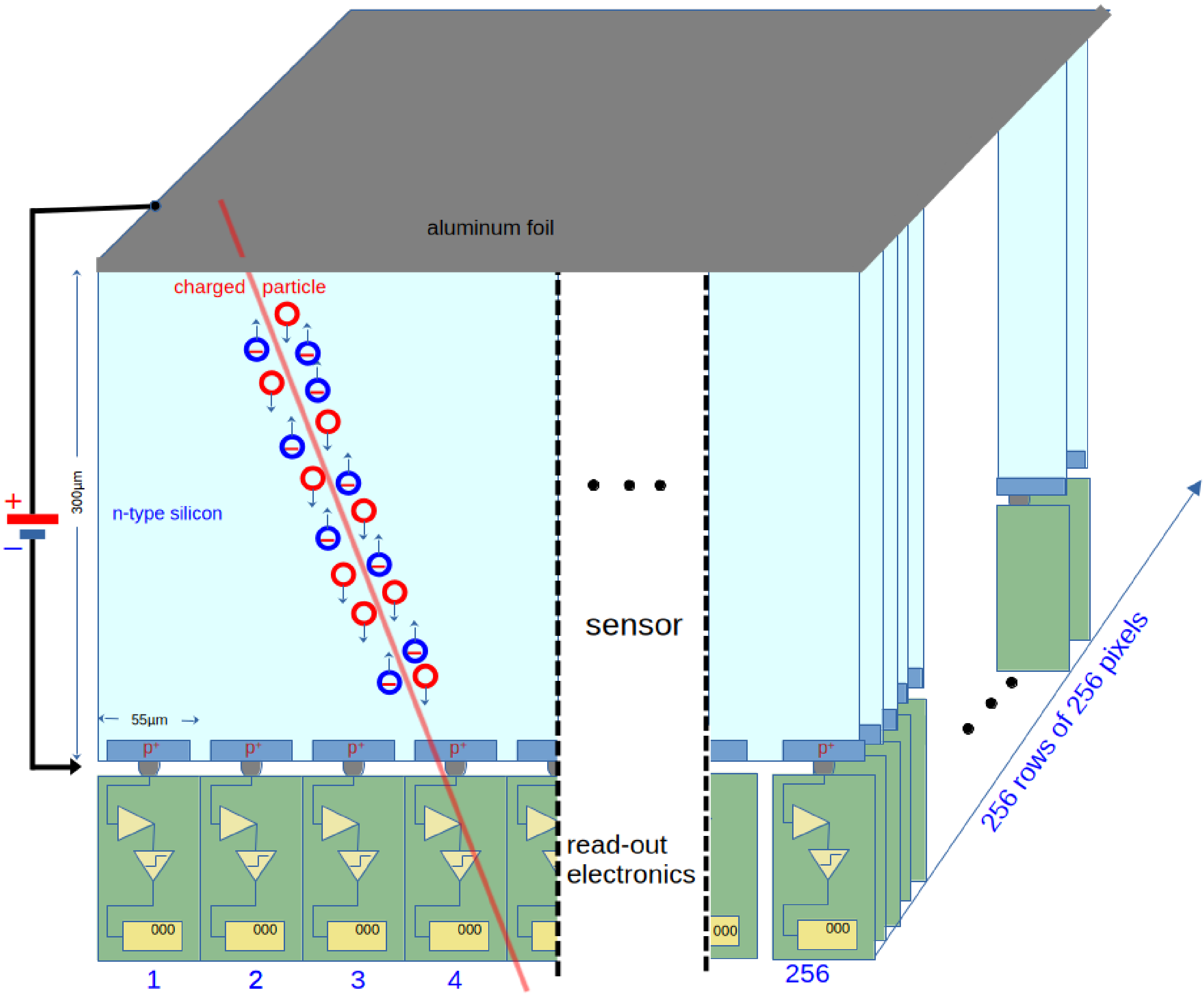


Abbildung 2: Schematic layout of the miniPIX detector

- studies of photon interactions in matter (dominated by the Compton process),
- energy loss of muons and its fluctuations along the trace.

There are even more benefits related to the practical application in laboratory courses or school experiments:

- use of low-activity radioactive samples to study natural radioactivity in hands-on experiments by students,
- freely adjustable exposure time to adapt to different radiation intensities,
- negligible noise rates, efficient suppression of backgrounds and well-defined detection lifetime,
- digital overlay of many recorded frames to obtain feature-rich images even from low-activity sources,
- storage of data to disk for later, in-depth analysis of particle signatures,
- different levels of pre-processing of recorded frames to adjust to the students' level of knowledge,
- usage of the very same detector for many different types of measurements.

In particular the last item is important, as, with the *miniPIX*, the properties of radiation itself are brought into focus rather than the peculiarities of the classical arsenal of detection techniques like electrometer, ionization chamber, Geiger counter, scintillation counter or mono-crystal silicon detectors.

As an example, the graphical display of a data collection run with *mPIXdaq* is shown below with an overlay of 10 frames, each recorded with an exposure time of 1 s. Clustering of connected pixels and classification of the patterns are performed in real time during data acquisition and results are shown as animated histograms. The history over the last 300 frames of the numbers of observed objects per frame is also displayed and demonstrates the Poisson nature of the underlying processes of production and detection of radioactive particles. More details on the real-time analysis method will be given below.

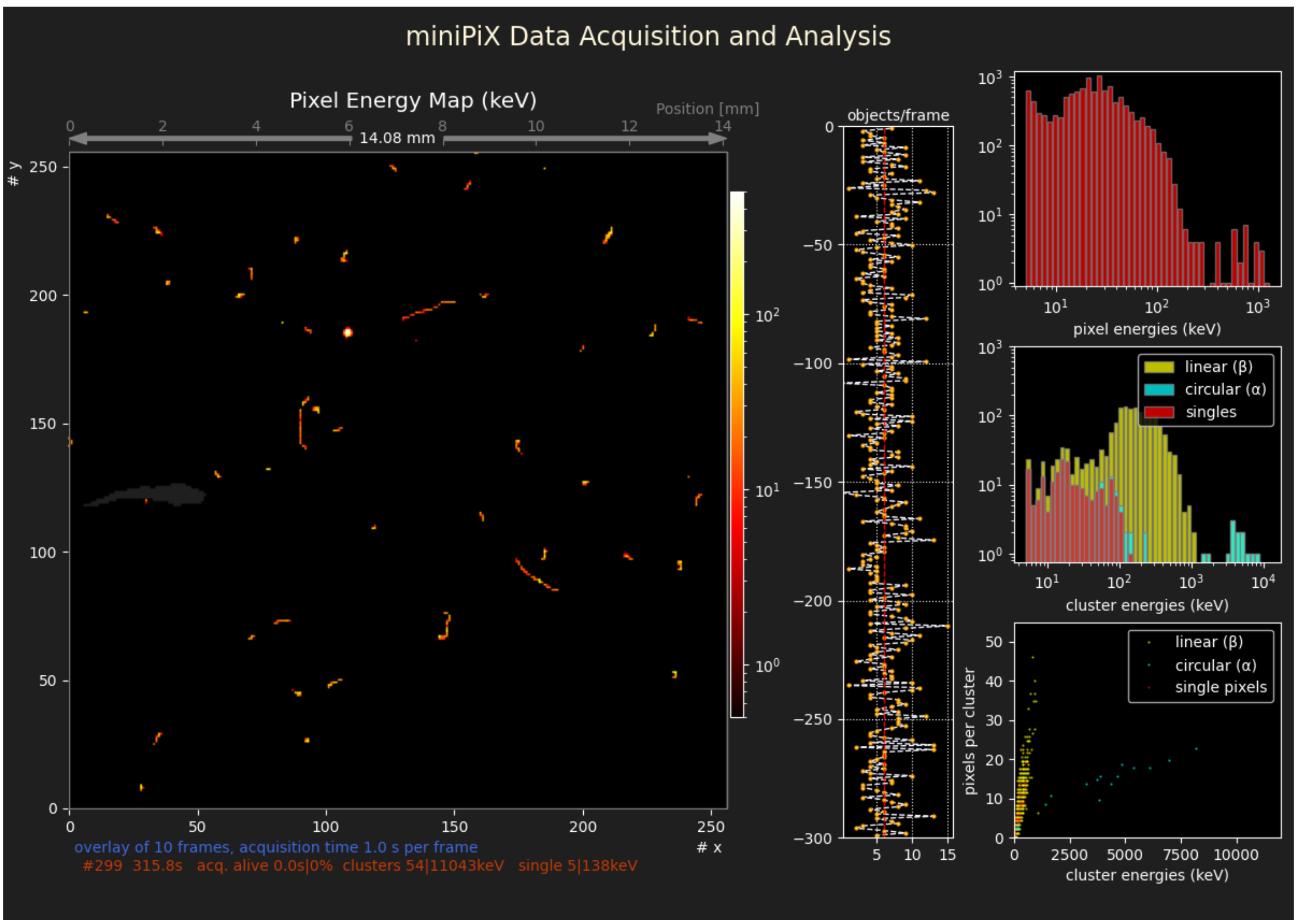


Abbildung 3: Pixel map and histograms of a Columbit sample recorded with *mPIXdaq*

Energy spectra of γ rays cannot be measured with the *miniPIX* owing to the small overall sensor volume which is not large enough to contain all energy deposits. Therefore a gamma spectrometer with a scintillating crystal is recommended for γ spectroscopy, e.g. one of the very cost-effective and sufficiently precise devices made by RadiaCode.

Examples of successfully conducted measurements in the student labs at the Faculty of Physics at Karlsruhe Institute of Physics are presented and discussed below.

Analysis of radiation from natural samples

The high sensitivity, combined with the ability to clearly distinguish different types of radiation, offers to take a new approach to introducing the topic of radioactivity. Radiation from low-activity, natural sources is sufficient to identify typical signatures and relate them to the different properties concerning their interaction and with matter.

Here, a small sample of natural Pitchblende (Uraninit, Uranium Dioxide) is chosen as the entry point for studies of radioactive phenomena. A typical *miniPIX* frame image recorded with *mPIXdaq* is shown in the figure below. Pixel energies are encoded as colors on a logarithmic scale, as indicated by the legend on the right-hand side of the image.

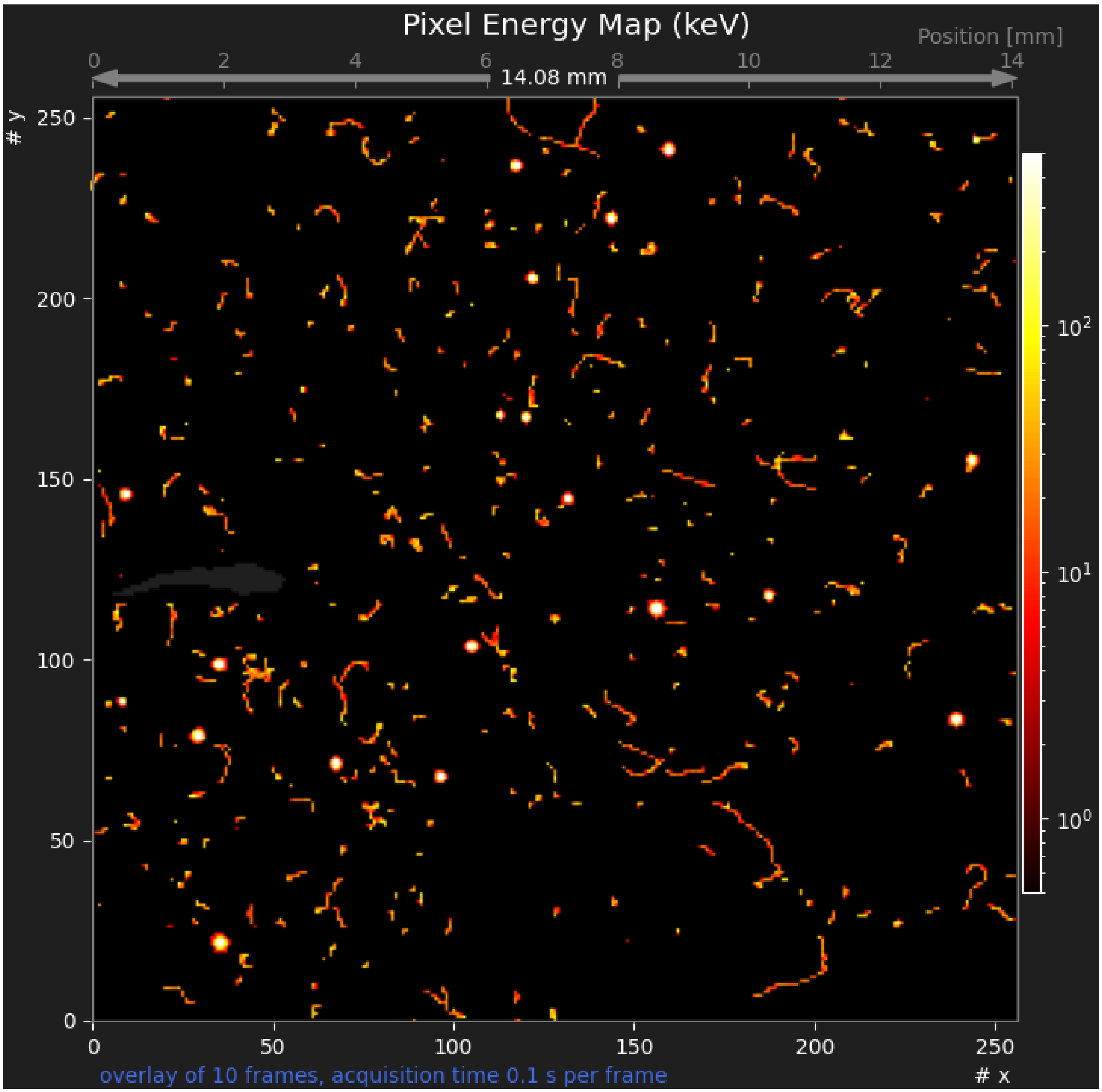


Abbildung 4: Display of *mPIXdaq* for Pitchblende

Three types of signatures are clearly distinguishable: circular “blobs” from α particles, long “worms” from β particles, and typically small objects with only very few pixels from energy transfers of γ rays to electrons in the silicon.

Enlarged views of typical α, β and γ signatures show this clearly. Note that the lengths of β traces depend on their energies and incident angles; usually, they are not fully contained within the sensitive volume of the *miniPIX* sensor. γ rays typically transfer only a fraction of their energy to electrons via the Compton process and therefore lead to signatures with only one or very view active sensor pixels. Note that lager energy transfers and the full transfer of the γ energy to electrons via photo-effect are also possible in rare cases.

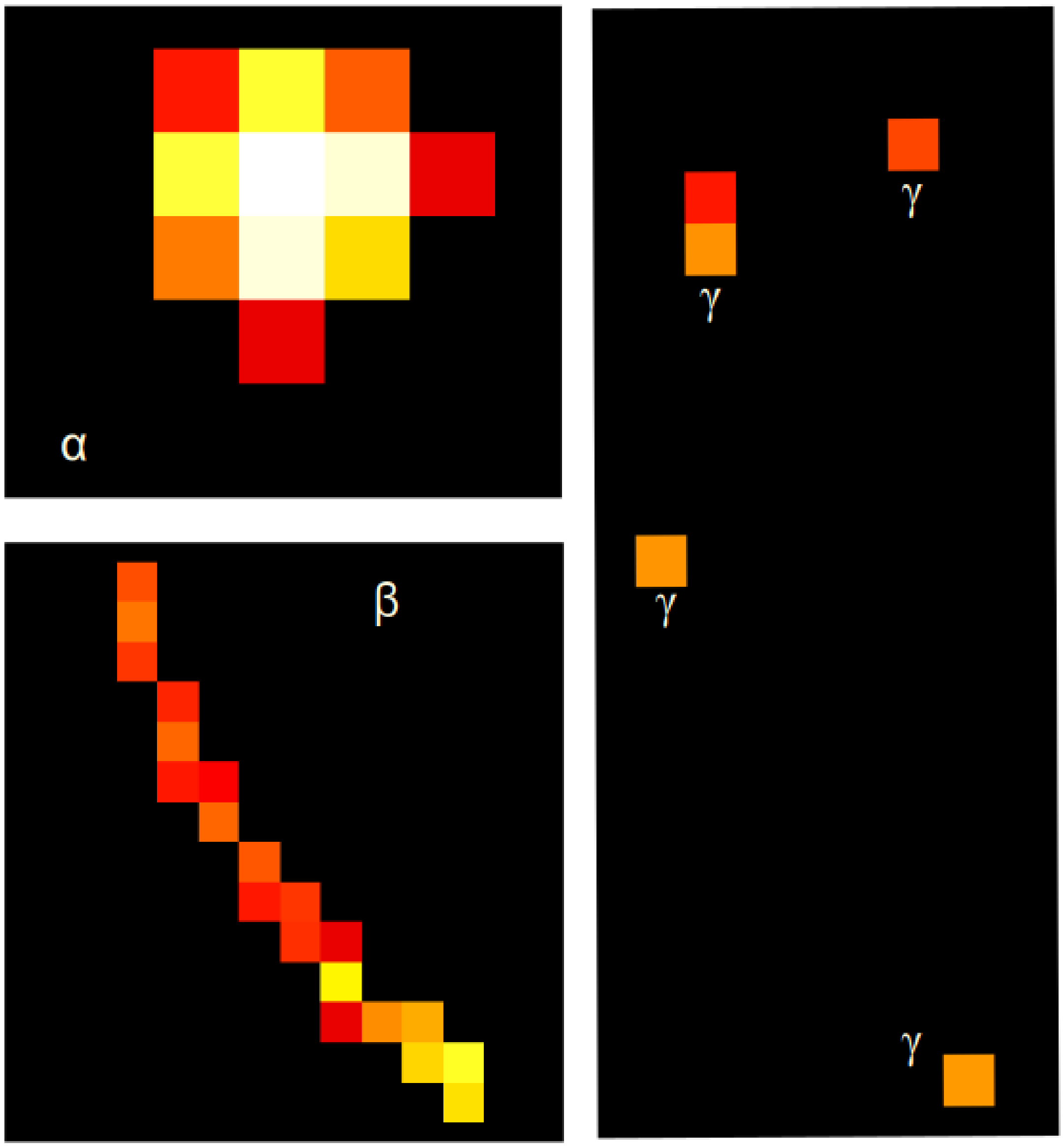


Abbildung 5: α, β and γ signatures in *miniPIX* frames

A **thin plastic foil** as absorber leads to complete suppression of all α and of low-energy β signatures. As becomes clear from the image shown below, the rate of recorded objects is significantly reduced, and the typical signatures from α particles are completely missing.

A **3 mm aluminum absorber** is sufficient to completely suppress all α and β particles such that only γ rays reach the sensor. A typical image is shown below. All traces are produced by γ interactions in or near the sensitive volume of

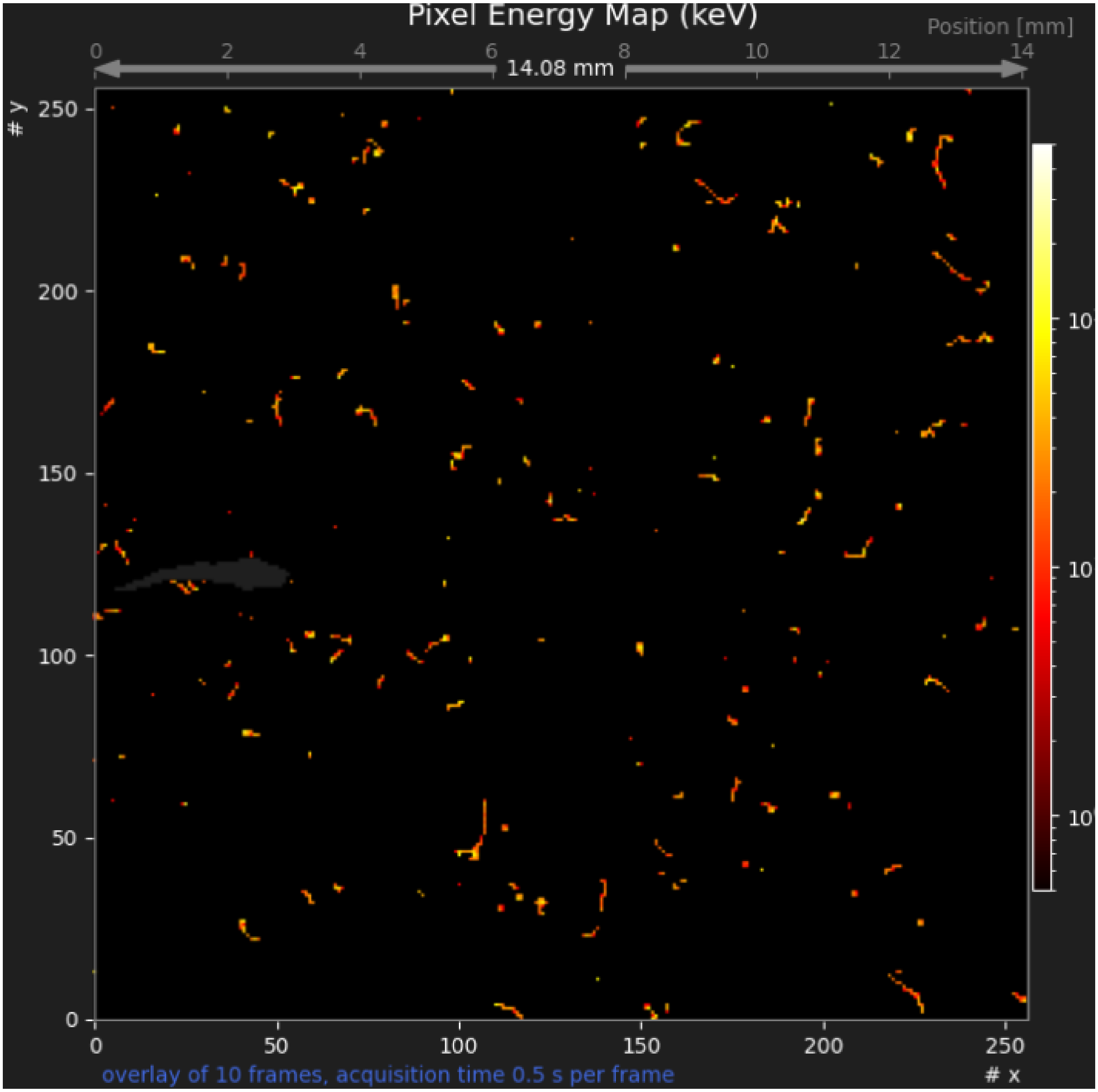


Abbildung 6: Display of miniPIXdaq for Pitchblende with thin plastic absorber

the detector; many involve only one or very few pixels. Longer traces occur when a significant amount of energy is transferred to electrons.

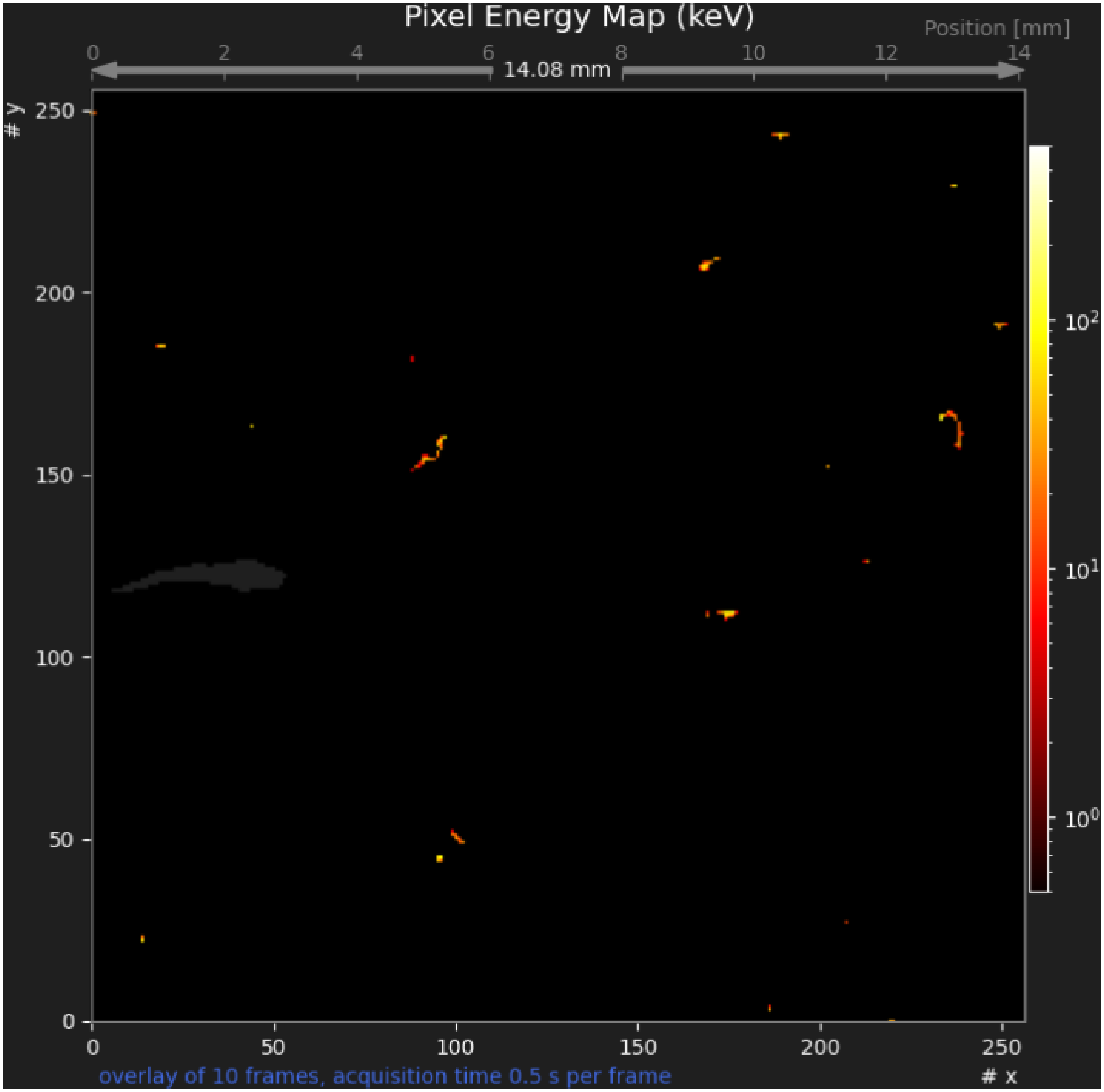


Abbildung 7: Display of miniPIXdaq for Pitchblende with Aluminum absorber

This sequence of images nicely demonstrates many features of radioactivity:

- there are three types of very distinct signatures of radioactivity from natural sources: very intensive circular patterns (α), long traces (β) and low energy deposits with only one or very few active pixels (γ),
- α and β particles can easily be shielded,
- γ rays are much more penetrating and more difficult to shield,
- signatures from γ rays look very similar to those of low-energy electrons, leading to the conclusion that photons interact with matter by transferring their energy to electrons that finally produce free charges (or electron-hole pairs in a semi-conductor).

Appropriate radioactive samples with activities of some 10 Bq are freely available and can be bought from educational supply stores, e.g. NTL.

Pedagogical considerations

From a pedagogical point of view, it is very appealing to start a course on radioactivity from experimental observations of a natural phenomenon and let pupils or students discover the properties of radiation by themselves.
In such an approach, the historical context becomes less important, as well as the sequence of developments of technological methods for radiation detection. Utilizing modern detection techniques right at the beginning puts emphasis on the phenomenon of radioactivity itself and on the interaction of radiation with materials.
Furthermore, the high sensitivity of the miniPIX detector and its ability to clearly discriminate different types of radiation avoids the use of artificial, high-activity radioactive sources to produce signals exceeding background counts. With the *miniPIX* device, it is possible to perform background-free counting and energy determination of α particles, or counting of β particles and γ rays with low-activity sources.
This approach permits studies with sources of much lower activities than needed in classical experiments e.g. to study the absorption of α particles in air.
In contrast, with a classical radiation detector, e.g. a Geiger-Müller counter, only a reduction in counting rate is observed when adding absorbers, but no distinction of the origin of the "clicks". Therefore, a high-rate α-source is needed to demonstrate complete absorption, with the caveat that unavoidable background counts from γ rays are still present.

Images like the ones shown above may be produced using the *mPIXdaq* program, or also with the *Pixet* (basic) program the vendor provides together with the *minPIX* detector. *mPIXdaq*, however, provides a transparent algorithm for clustering of pixels and for the characterization of cluster properties that rely solely on methods that are mastered already by undergraduate students. *mPIXdaq* allows highly selective recording of special cluster types, thus strongly discriminating the desired signatures against backgrounds. The *miniPIX* data may also serve as a motivation for younger high-school students to learn complex methods of digital data processing and analysis.

The algorithms are fast enough to be deployed on-line in low-rate scenarios, thus providing quantitative results on particle rates and energy spectra. An example of the energy spectra of linear and circular clusters and of single pixels is shown below.

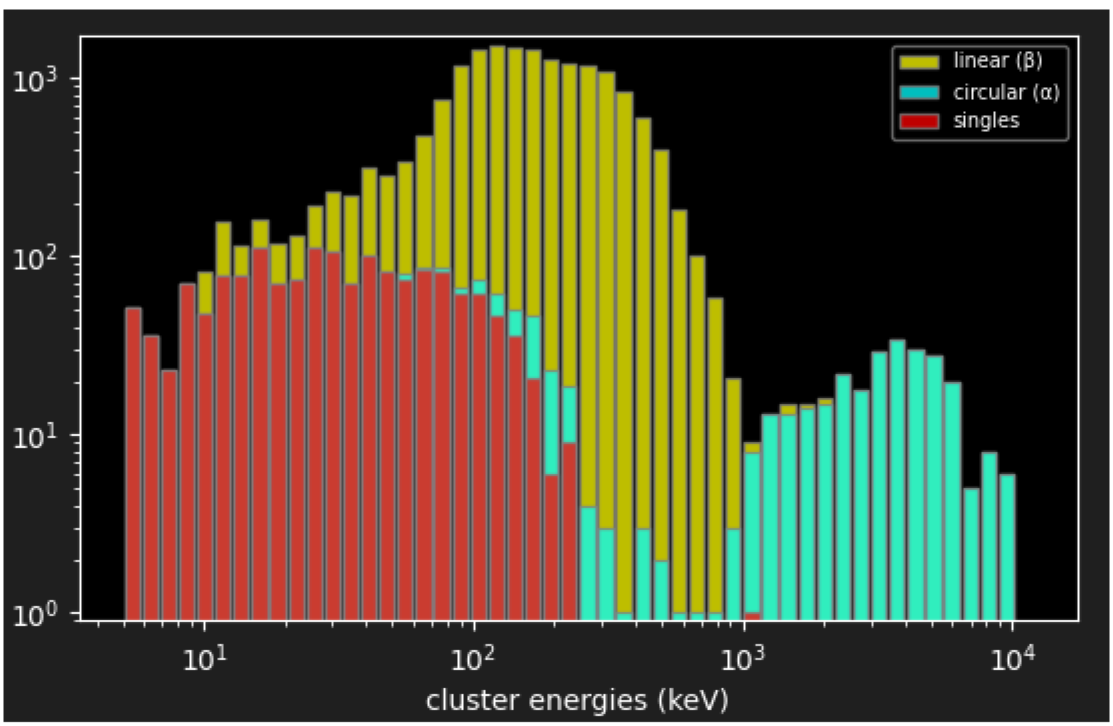


Abbildung 8: Energy spectra determined with miniPIXdaq

Real-time analysis of recorded data

mPIXdaq performs an online analysis of recorded data frames and presents the results as animated histograms which are updated as data collection proceeds. Raw or clustered frame data as well as cluster properties in *.csv* format can be created with *mPIXdaq* and stored on disk for subsequent analysis. In addition to the very beneficial visual impression it thus becomes possible to use computer-based methods to study in detail the properties of energy depositions by different types of particles.

In a first step, connected areas of pixels, called clusters, are determined in each recorded frame using the *label()* method of the image-processing library *scipy.ndimage*.

The main features of each cluster are the mean position in pixel coordinates, the number of pixels,the sum of all pixel energies and the maximum energy in a single pixel of the cluster. The postion and size of a cluster is approximated by the “bounding box”, a rectangular area than contains all pixels of the cluster. A α particle activates most pixels in the box, i.e. the number of pixels is given by the product of the width, *w*, and height, *h*, of the bounding box. In contrast, β particles only activate pixels along their trace, and the minimum number of active pixels corresponds to the length of the diagonal of the box.

Further interesting cluster features are the geometrical shape of the cluster area and the shape of the energy distribution over the pixels. To characterize the geometrical shape, the covariance matrix of the pixel coordinates, cov(x_i, y_i) is used. Stored are the half-lengths of the principal axes (or the semi-major and semi-minor axes) and the angular orientation of the principal axis of the covariance ellipses of the clusters. Almost identical values of the half-lengths classify a circular geometry, while largely different values are characteristic of liner topologies. The ratio of the two half-length it therefore used as a measure of the “circularity” of the cluster, which already provides a good separation of α and β particles.

A further, very sensitive variable is the covariance matrix of the energy distribution in the clusters, cov$(E(x_i, y_i))$. For α particles, this distribution peaks at the centre and steeply falls off towards the boundaries, leading to a small variance. The properties of the covariance ellipses of the energy distribution are stored analogous to those of the geometrical ellipses.
A small ratio of the lengths of the semi-major axes of the two ellipses, called “flatness”, is a very prominent signature of α particles.

Optionally, in addition to the cluster properties, a list of pixels contributing to each cluster can be stored for more sophisticated off-line analysis.

As a starting point for own analyses, the *mPIXdaq* package offers a *Jupyter Notebook*, *analyze_mPIXclusters.ipynb*, for use with a local or remote *Jupyter* service. In standard *Python* environments, such a server can easily be set-up, as is documented on the project homepage *jupyter .org*.
The analysis example provided as part of the *mPIXdaq* package shows how to read the output files and also provides a sample analysis. The code relies on the *pandas* package which has become a well-established standard in data science for the analysis of large datasets.

The following cluster features are derived by *mPIXdaq* for each pixel cluster during online-processing:

[‘time’, ‘x_mean’, ‘y_mean’, ‘n_pix’, ‘energy’, ‘e_mx’, ‘x_mn’, ‘y_mn’, ‘w’, ‘h’,
‘var_mx’, ‘var_mx’, ‘var_mn’, ‘angle’, ‘xE_mean’, ‘yE_mean’, ‘varE_mx’, ‘varE_mn’]

```
time    : time since start of daq when frame was recorded
x_mean  : mean x-position of cluster (in pixel numbers)
y_mean  : mean y-position of cluster (in pixel numbers)
n_pix   : number of pixels in cluster
energy  : energy of cluster (= sum of pixel energies) in keV
e_mx    : maximum pixel energy
x_mn    : minimum x of rectangular bounding box containing the cluster
y_mn    : minimum y of bounding box
w       : width of bounding box
h       : height of bounding box
var_mx  : maximum variance of geometrical cluster shape (in pixels)
var_mn  : minimum variance of geometrical cluster shape (in pixels)
angle   : orientation of cluster (0 = along x-axis, pi/2 = along y-axis)
xE_mean : mean x of energy distribution  (in pixel numbers)
```

```
yE_mean : mean y of energy distribution  (in pixel numbers)
varE_mx : maximum variance of energy distribution
varE_mn : minimum variance of energy distribution
```

This set of variables already permits very detailed studies of the properties of energy deposits in the *miniPIX* detector. An almost perfect separation of the different types of radiation becomes possible and permits background-free selections of α and β traces, counting their rates and determining energy spectra of α particles.

As a further option, a file with cluster data stored in *yaml* format contains the positions and energies of all contributing pixels. More sophisticated analysis strategies can thus be explored. As an example, it is possible to identify β tracks that are stopped in the active volume of the detector by using the large enhancement of the deposited energy when the electrons come to rest in the material. Explicitly selecting such tracks opens up some limited possibilities for β spectroscopy.

Advanced topics for university lab courses

Typical (classical) experiments in nuclear physics, like energy spectra and energy loss in air of α particles or the penetration power and absorption of β radiation can comfortably be performed with the *miniPX* detector. Its granular spatial resolution allows further insights to be gained that are not achievable otherwise.

To support quantitative experiments, the *Python* script `calculate_dEdx.py` for the determination of the (mean) specific energy losses (dE/dx) of electrons and alpha-particles in air and silicon is provided with this package. The calculated energy loss of particles (or, resp. the deposited energy in the absorbing material) are important to relate the measured signals to theoretical expectations.
The calculated energy deposits are based on modified versions of the Bethe-Bloch relation for the energy loss of charged particles in matter and represent reasonable approximations. The presently most authoritative information source on energy losses of electrons, photons and Helium nuclei (α particles) are the tabulated data by NIST (ESTAR, ASTAR and PStar), see *NIST Standard Reference Database*.

A selection of experiments with the *miniPIX* detector and the *mPIXdaq* package is described in the sub-chapters below.

Penetration depth of α particles in air

The **penetration depth** of α particles and the energy loss in air can be directly determined with the *miniPIX* by replacing any other detector in an existing setup. With the *mPIXdaq* package, signatures of α particles are identified without any backgrounds and their energies are measured with sufficient precision to observe the shift in energies as a function of the distance between the source and the detector.

The expected behavior of the measured α energies as a function of the depth of penetrated air, as determined with *calculate_dEdx.py*, is shown in the figure below.

The energy loss per 0.5 mm traversed length of air, equivalent to the deposited energy, is shown in red; it rises by almost a factor of four at the end of the α reach when the particles become very slow. This behavior illustrates the Bragg peak of the deposited energy, which is relevant for radiation therapy.

Quantitative measurements to demonstrate the energy loss in air (or the “stopping power) can be made if a thin, preferably open and mono-energetic α-source, e.g. Am-241, is available. α particles are selected using the cluster properties explained above by requiring a high value of the average energy deposit per pixel, a circular cluster shape and a non-flat energy distribution.
Here is a code example, assuming the data is stored in a *pandas* data frame *df*:

```
# define useful quantities
# - mean energy per pixel
df['Epix_mean'] = df['energy'] / df['n_pix']
# - circularity of cluster shape as the ratio of the sizes in x and y
df['circularity'] = df['var_mn'] / np.maximum(df['var_mx'].to_numpy(), 0.001)
# - flatness of energy distribution
df['flatness'] = df['varE_mx'] / np.maximum(df['var_mx'].to_numpy(), 0.001)

# apply cuts on these quantities
```

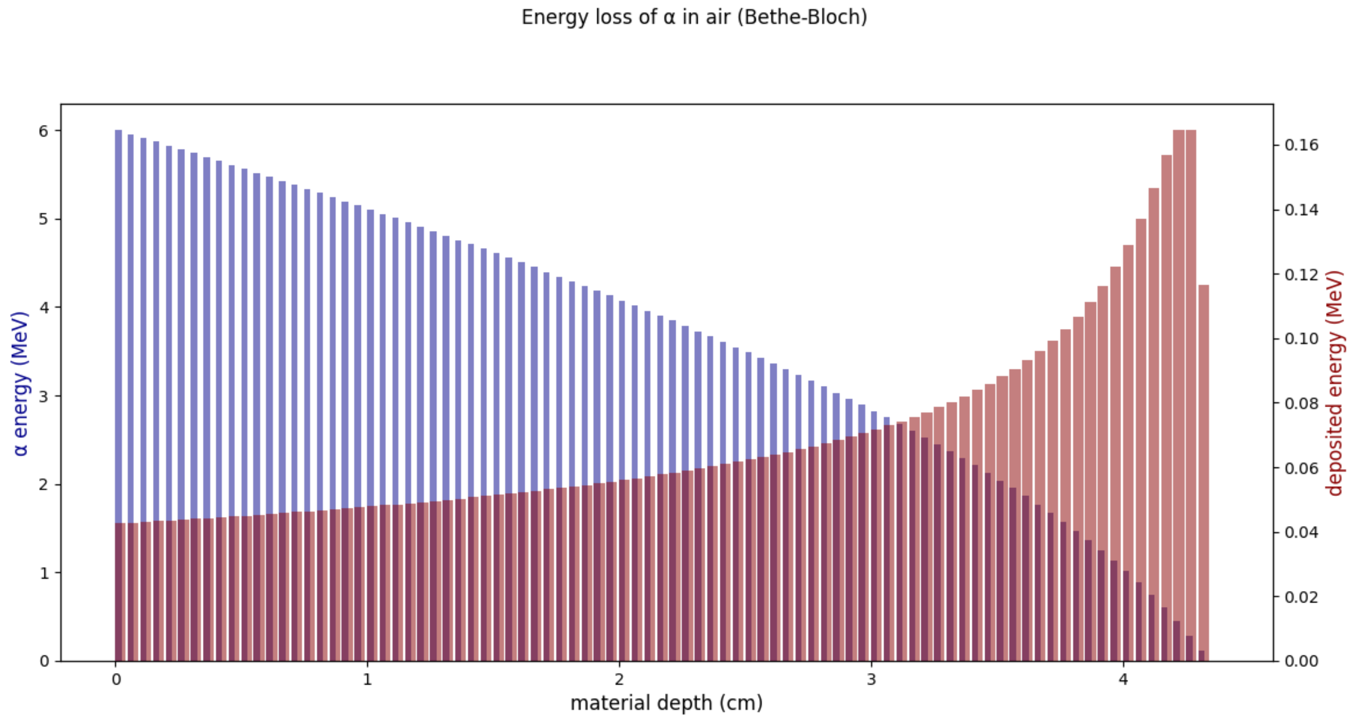


Abbildung 9: α energy as a function of the penetration depth in air

```
dEdx_cut = 80
circularity_cut = 0.3
flatness_cut = 0.4
is_high_dEdx = df['Epix_mean'] > dEdx_cut
is_circular = df['circularity'] >= circularity_cut
is_flat = df['flatness'] > flatness_cut

# loose definition of alpha candidate by shape
shape_is_alpha = is_circular & ~is_flat
# a loose definition of an alpha
is_cand_alpha = shape_is_alpha | is_high_dEdx
# a tight definition of an alpha
is_alpha = shape_is_alpha & is_high_dEdx
```

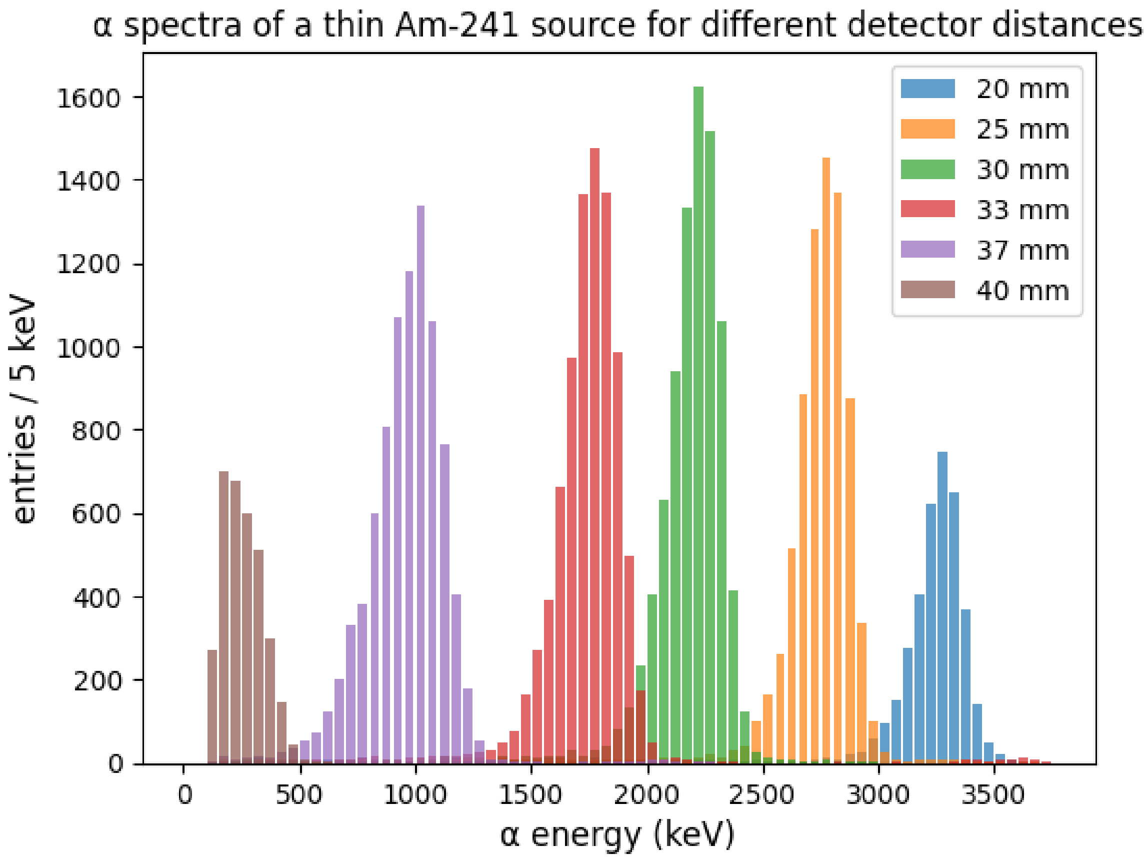
To obtain reliable energy measurements, it is important to exclude clusters with pixel energies above 1200 keV, because the detector response becomes non-linear due to saturating pixel charges. In fact, the energy is largely overestimated in such cases. A cure is to discard events where α particles hit the centre of a pixel, leading to the definition of a “clean alpha” with reliable energy determination:

```
is_clean_alpha = is_alpha & (df['e_mx'] < 1200.)
```

or, if also low-energy α with one to three active pixels are to be included:

```
is_clean_alpha = is_cand_alpha & (df['e_mx'] < 1200.)
```

An example of such measurements with a thin, open Am-241 source and the *miniPIX* placed at varying distances away from from the source is shown below. The energy resolution (full width half maximum) is of the order of 5%.



The distance between the peaks, i.e. the energy loss in air, rises at lower energies, as expected. The energy loss amounts about 100 keV/mm at an energy of 3 MeV and increases to about 200 keV/mm at 1 MeV.

The mean energies obtained from the fitted peak positions as a function of the distance of the *miniPIX* from the Am-241 source, together with theoretical expectations from the Bethe-Bloch formula, are shown below. The vertical error bars indicate the standard deviation of the spectra; the uncertainty on the mean energy is very small and not visible.

The observed range of 4.1 cm for α particles from Am-241 with an energy of 5.5 MeV corresponds well to the prediction. There are, however, deviations from the expectation at smaller distances, which are most likely caused by the known non-linear behavior of the *miniPIX* at high α energies:

- The cut on the maximum energy in a single pixel biases the determined energy towards lower values.
- The layer of dead material in front of the sensitive area of the *Timepix* chip leads to a small energy-dependent correction on the order of 100-200 keV, corresponding to the energy loss in 1 mm of air.

- The energy loss depends on the air density and hence on temperature and air pressure. The dashed and dashed-dotted curves show this effect for variations of the air density of ±5%, the size of typical variations under different weather conditions.

A careful calibration of the *miniPIX* for high α energies is needed if more precise studies at high α energies > 2 MeV are desired.

Measurement of the energy loss of β radiation.

Absorption curves are determined by measuring the rate of β traces as a function of the thicknesses of absorber material placed between a β source (typically Sr-29/Yr-90) and the *miniPIX* detector. As in experiments with classical detectors, these are pure rate measurements, because β traces are not fully absorbed in the sensitive volume of the *miniPIX* and a measurement of their energy is only possible if the traces are fully contained, i.e. for energies below 200 keV.

The quantity of interest in such measurements is the mass absorption (or attenuation) coefficient, μ/ρ in units of cm²/g. It is obtained from the dependence of the count rate on the traversed material thickness. Typically, the absorber

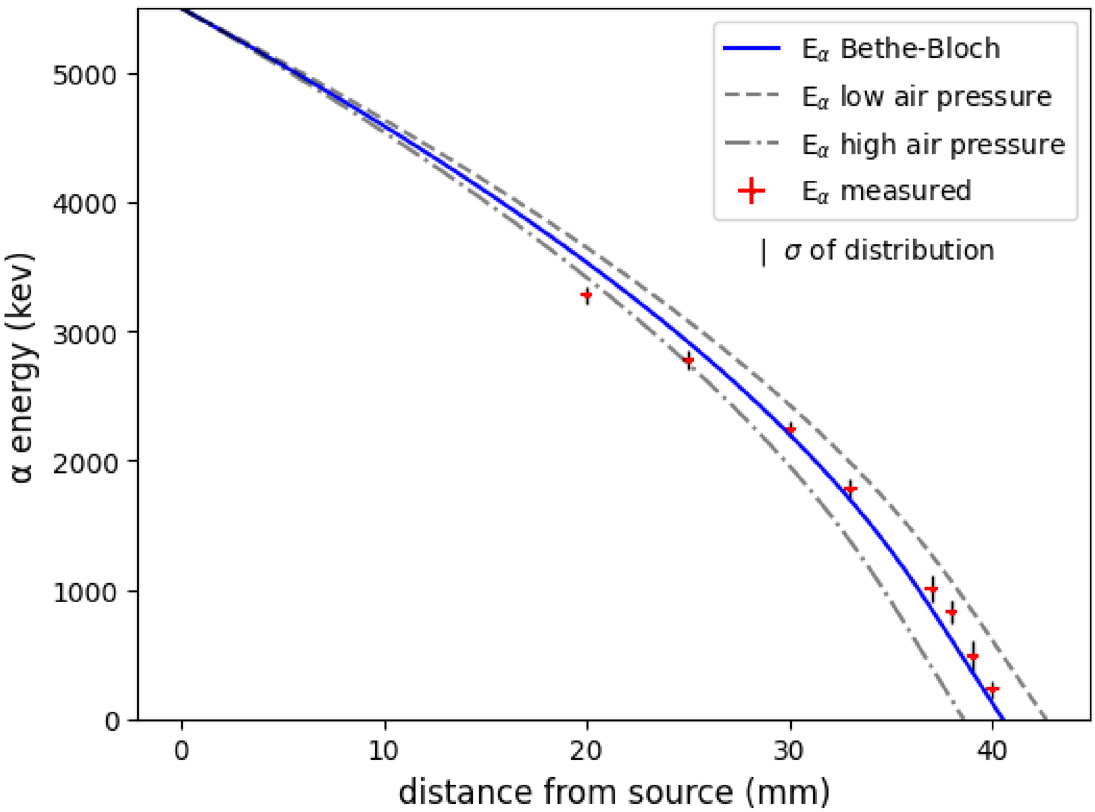


Abbildung 10: α energy as a function of the penetration depth in air

material consists of very thin aluminum foils of some ten µm thickness.

In this experiment, the measured rates depend on the energy spectrum of β radiation emitted by the source, which is folded with the absorption properties of the material.

Fortunately, these two effects can be disentangled with the *miniPIX* device, as will be shown next.

Absorption in Silicon The energy loss of β radiation in Silicon can be directly studied with the *miniPIX*, because the pixels act as absorber and detection material at the same time. The energies deposited in the pixels along a β track directly show the energy loss (dE/dx) along the trace. Because the β energy decreases while traversing the silicon by exactly the measured energy in the pixels, the recorded energy in each pixel represents a measurement of the energy dependence of the specific ionization loss in silicon.

The expected mean energy loss per pixel, as calculated with *calculate_dEdx.py* using a modified Bethe Formula, shows a strong increase at the end of the tracks where the electrons become slow. Predicted is an average energy deposition of about 20 keV for electrons with energies between 0.2 and 1.5 MeV, rising to some ten keV for electrons with kinetic energies below 200 keV.

An example of such a long trace of a β particle is shown in the pixel energy map below. Note that fluctuations of the energy deposits around the mean are large for thin layers of absorbing material. The energy deposits in each pixel are very close to the theoretical expectations.

The image was produced with the *Jupyter* notebook *analyze_mPIXclusters* and the function *plot_cluster()* from *mpix-daq.mpixhelpers* by selecting non-α tracks with a large number of pixels. The particle enters the detector from the top-left and then loses energy in each pixel as it traverses the sensitive silicon volume. It finally stops in the lower-left corner, where the energy deposit per pixel shows the expected increase.

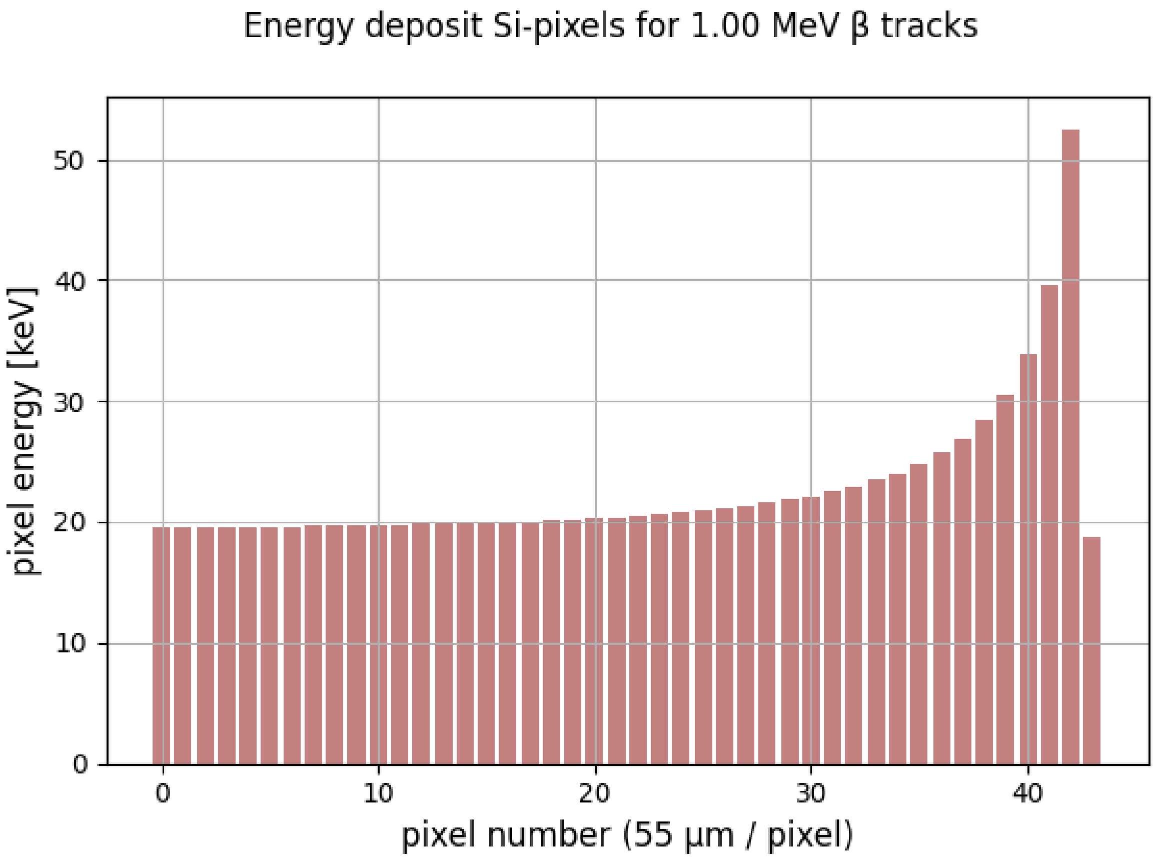


Abbildung 11: Energy deposit per pixel for β particles

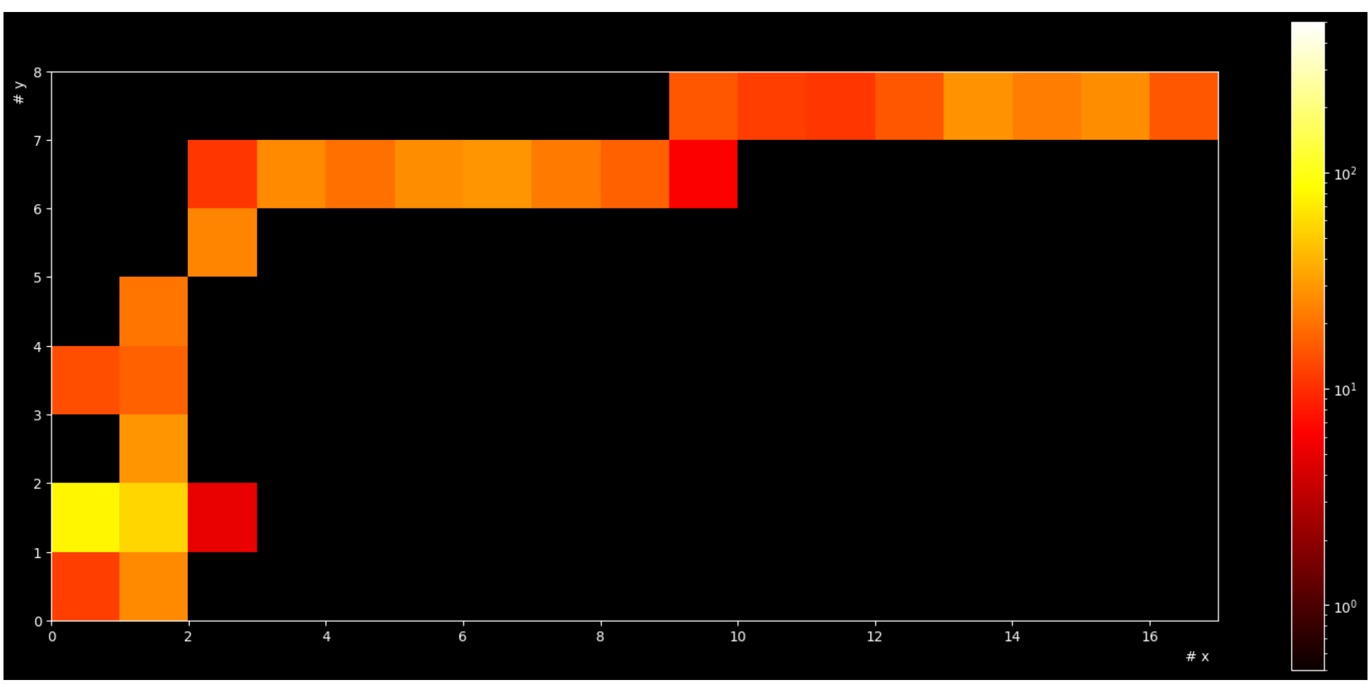


Abbildung 12: Long β track

In a more comprehensive analysis, the electron energy in a given pixel is obtained by adding up all energy losses starting from the stopping-point. If many such tracks are sampled, it becomes possible to determine the energy loss per pixel as a function of the electron energy. As can be seen from the electron track above, the energy deposit per pixel is not constant, but shows large, non-Gaussian fluctuations, which are approximately described by a Landau distribution exhibiting a long tail towards large energy deposits due to rare, but intense interactions. It is also important to note that the path of a track in a pixel is not precisely known, because it may cross the sensitive area at an unknown elevation angle w.r.t the pixel plane. Furthermore, traces traversing the pixels not in the x- or y-directions have a longer path length by up to a factor of $\sqrt{2}$, and the energy of tracks passing at the pixel edges my be shared between adjacent pixels. With appropriate selection of tracks, these problems can be mitigated. Long tracks have a small inclination angle with the pixel plane, and neighboring pixels can be summed up by projection them on the x or y axes for tracks oriented along the x- or y-direction, respectively.

Data were recorded with a Strontium-90 source by reading 10000 frames with an exposure time of 50 ms, resulting in a large data set of 350000 clusters in total. From these, 680 tracks were selected that satisfy the strict selection criteria described above. The stopping point of the tracks are identified by requiring an energy deposit larger than 70 keV in the last two pixels. The mean deposited energies per pixel for long β tracks traversing the pixel plane under an angle not larger than 22.5° from the x- or y-directions are shown in the graph below. The pixel where the track ends is labeled as 1.

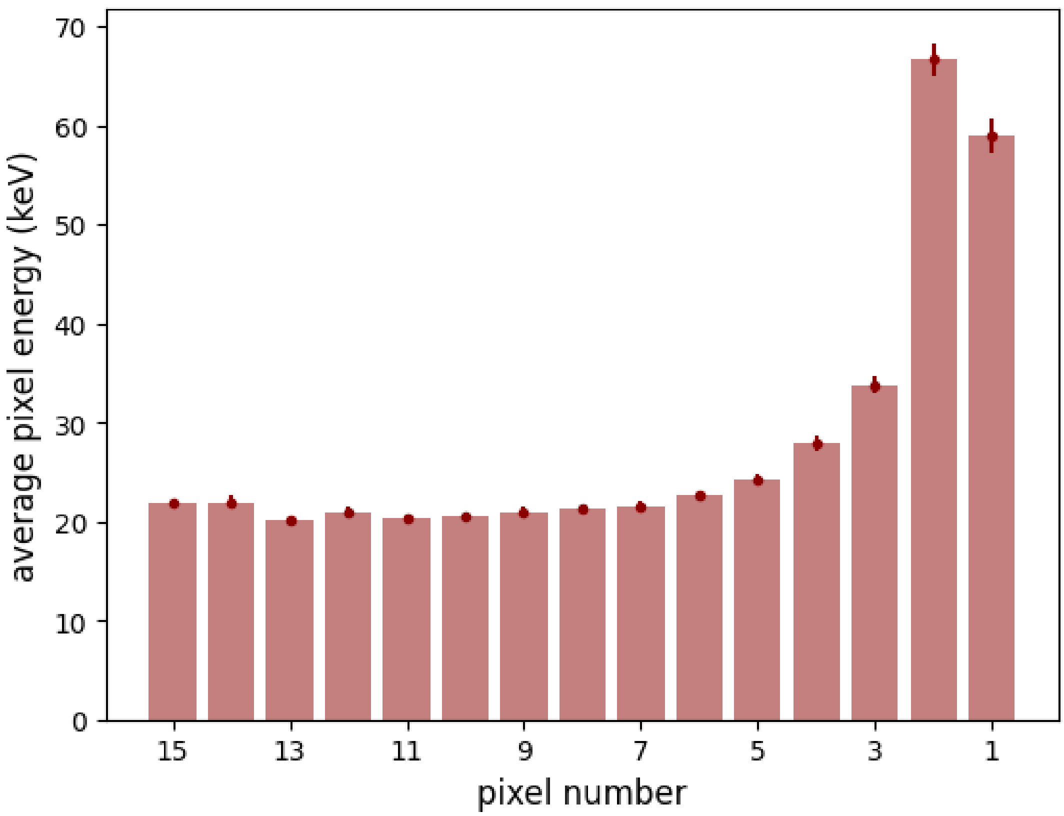


Abbildung 13: Measured mean energy deposit per pixel for electron tracks before stopping point

The measured mean energies correspond well to the theoretical expectation discussed earlier. The energy deposits becomes smaller with increasing distance to the stopping point, i.e. with rising track energy. The value in pixel 15 is 21.8 keV, which also corresponds well to the theoretical expectation.

The full energy distributions are shown for some pixels. They agree well with the Landau distributions fitted to the histograms, shown as dashed lines.

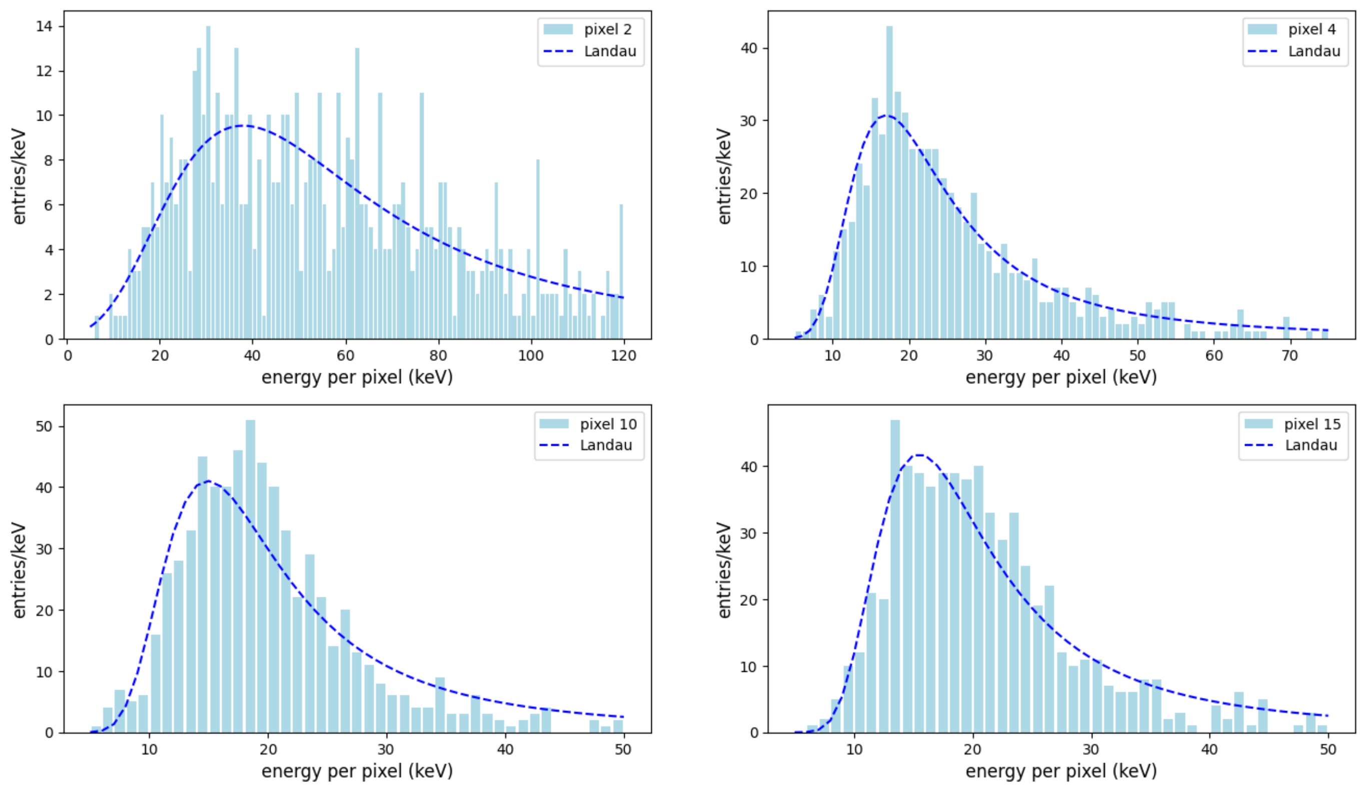


Abbildung 14: Energy deposit per pixel for electrons

Interactions of γ rays with silicon

Wit sufficient shielding, only γ rays reach the sensitive layer of the *miniPIX*. A collection of signatures produced by gamma interactions is shown below. The radioactive source was a low-activity stone from the Black Forest shielded with 3 mm of Aluminum. All signatures look like electrons at the end of their reach.

The energy spectrum ist typically very steeply falling, and any modelling strongly depends on the properties of ambient radiation under the given environmental conditions.

A clean source of photons of 59.5 keV is obtained by shielding the α particles of an Am-241 source. The energy spectrum of clusters with less than 5 pixels is shown in histogram below. The photo-peak at 59.5 keV is clearly visible, along with entries at lower-energy. The maximum energy of electrons produced by Compton scattering of the incident photons 19 keV is also clearly visible, along with other low-energy entries stemming from X-rays produced in the source or in the sensor material.

Ambient radiation

Ambient radiation at a normal level of 0.1μSv/h leads to an interaction rate of about 25 photon signals per minute in the active *miniPIX* volume of 0.059 cm³. This represents a respectable detection efficiency outperforming a small Geiger counter by a factor of two to three, and shows that the *miniPIX* can also be used for precision dosimetry. Still, this rate is low compared to typical count rates of 180 per minute observed in a CsI(Tl) crystal of one cm³ volume in a RadiaCode 102 device.

An overlay of 300 *miniPIX* frames with an exposure time of 1s each is shown in the figure below. This very feature-rich image shows clear signatures of photons. The rate of single pixels is about 0.5Hz - other signatures with a small number of pixels also originate from γ interactions, which dominate the rate. Other than possible with the above-mentioned scintillation counters, seven clear signatures of α particles and a large number of electron traces are also visible.

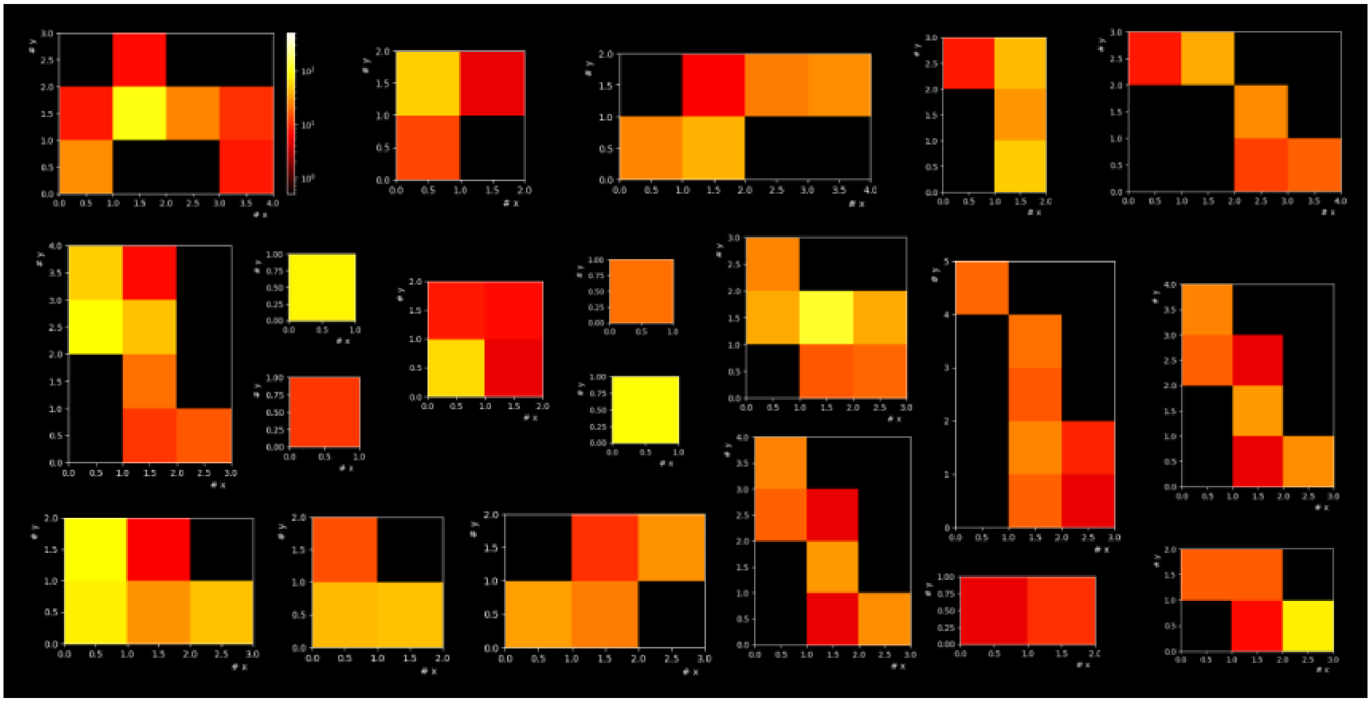


Abbildung 15: Energy deposits of electrons produced by γ rays

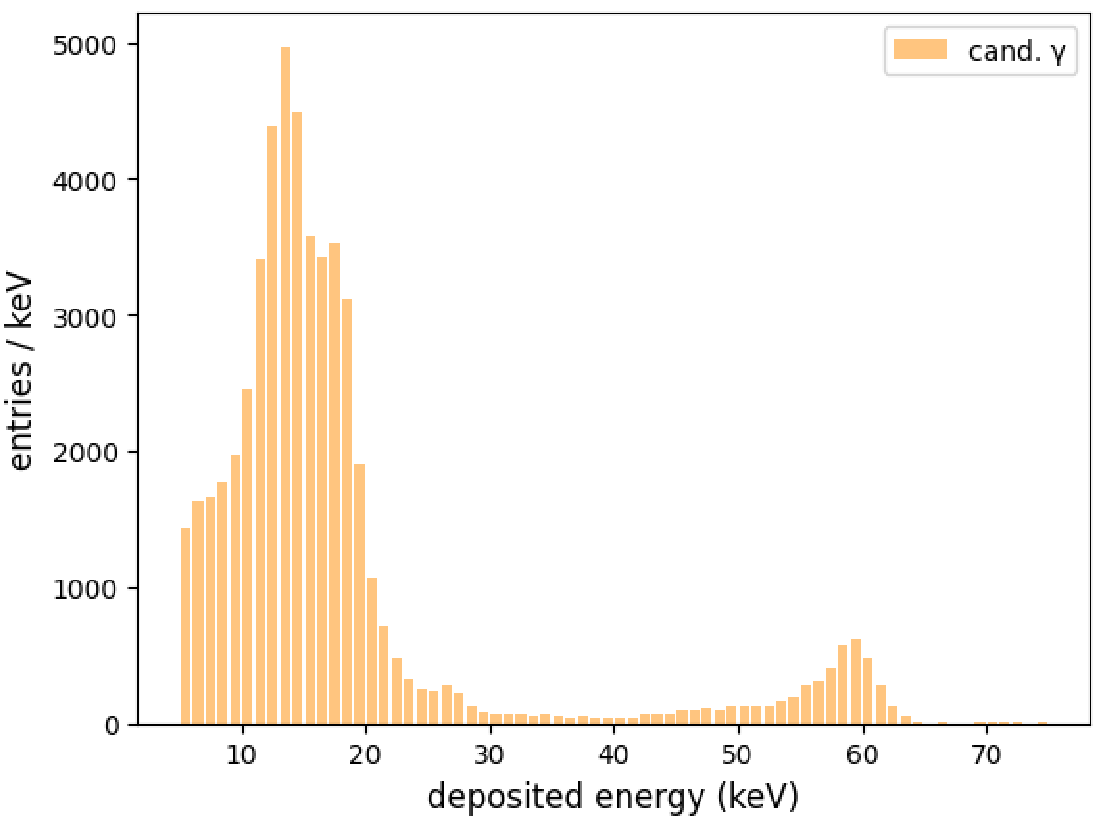


Abbildung 16: γ spectrum from shielded Americium

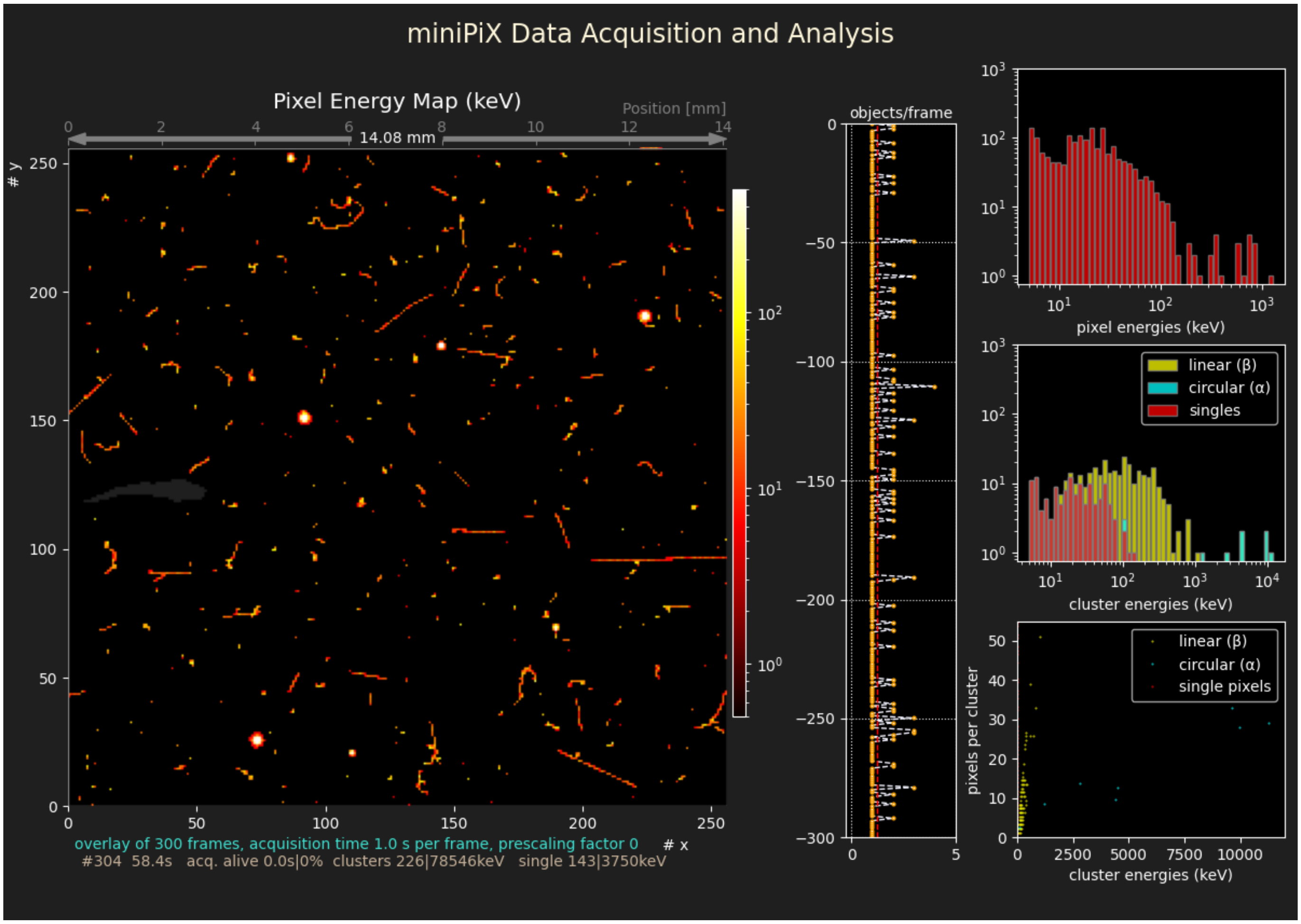


Abbildung 17: Ambient radiation

The α signatures stem from decays of the noble gas Radon (Rn-222 and Rn-224) from the Uranium and Thorium decay chains. Radon is produced from radioactive decays of these elements in building materials and in the inner of the Earth. As a non-reactive noble gas, Radon is released from the material and accumulates in the air, in particular in poorly ventilated rooms. Radon and its daughter-nuclei (Po, Bi, Pb) produce α particles with typical energies around 5 - 7 MeV. As they rapidly loose energy through interactions with air molecules the energies observed in the *miniPIX* are typically smaller.

The long, straight track near the centre at the right-hand side of the image is an example of a muon from cosmic radiation traversing the detector under a flat angle of about 10°. The sensor was oriented such that the *x*-axis pointed vertically.
Muons are heavy, and therefore they do not scatter much in the silicon, leading to very straight tracks. In addition, they are high-energetic and minimum-ionizing, and therefore their ionization loss along the track is constant. This is in contrast to electrons, which show a rise in ionization when they are slowed down in the material and lose a large fraction of their initial energy.
Tracks from muons can only be distinguished from electron signatures if the tracks are long enough, i.e. if they traverse the sensitive area under a rather flat angle so that many pixels respond. Note that a muon under 45° will fire only 5 pixels, and a muon under 30° 10 pixels. Most muons arrive under 90° from the top, and if the sensor is properly oriented, a noticeable fraction of the total muon flux is observable in the *miniPIX*.
With a sequence of measurements with different detector orientations studies of both the rate and direction of muons become possible. This will, however, require long measurement times because the expected rate of muons at sea level, integrated over all angles of incidence, is only about 1/cm²/min. Two clear muon tracks found in a data set recorded with a total acquisition time of 1000s are shown below.

The example of a recording of ambient radiation also illustrate the usefulness of the *miniPIX* as a **dosimeter** to monitor radioactive environments. Different conditions outdoors, in a well-ventilated room or in the typically badly ventilated basements of buildings are interesting locations to study.

Radon and its decay products can be easily accumulated on a paper towel with a vacuum cleaner or on the surface of an electrostatically chared balloon. An increase by a factor of ten compared to natural radiation levels for both α and β particles and for γ radiation is easily achievable.

An example of particle rates as a function of time from a paper towel that had been exposed to the air flow of a vacuum cleaner in the cellar of a building for 10 min is shown below.

Starting from a background rate of about 0.3 HZ, the rate of detected clusters or single pixels jumps to over 10 HZ when the sensor is exposed to the towel and then drops exponentially with time.

At the end of the Radon decay chain, Pb-210 with a half-life time of 22 years is produced, which, via a twofold β decay to Bi-210 and Po-210 with a subsequent α decay, finally reaches the stable Pb-206 nucleus. Pb-210 constitutes a significant source of natural environmental radiation and is enriched in areas with high Radon levels, like poorly ventilated cellars of buildings. This long-lived component is responsible for the flat part seen in the rate plot beyond 11000 s, which lies about a factor of three above the initial background level.
The short-lived nuclei produced from Radon decays offer one of the rare cases to directly observe an exponential decay law. Note, however, that this is not an exact exponential function because of the production of radioactive daughter nuclei which contribute to the recorded decay products.

Absorption of γ rays in materials

Studies of the absorption of γ rays in different materials as a function of the depth of traversed material and the initial γ energy are straight-forward with the *miniPIX* detector. With a set of γ sources, like shielded Am-241, Cs-137, Na-22 or Co-60, a sufficiently large variation of initial energies ranging from 60 to 1330 keV is available for such measurements. Absorber plates made of lead with thicknesses in the range from 1 - 25 mm or aluminum blocks of 15 - 25 mm thickness are also useful assets for such experiments.

The result of a sequence of measurements using a Cs-137 source with lead absorber plates is shown in the figure below. Contrary to what was seen for α spectra, the shape of the γ spectra remains almost unchanged when the thickness of the absorption layer changes. The rate however, decreases strongly with increasing absorber thickness.

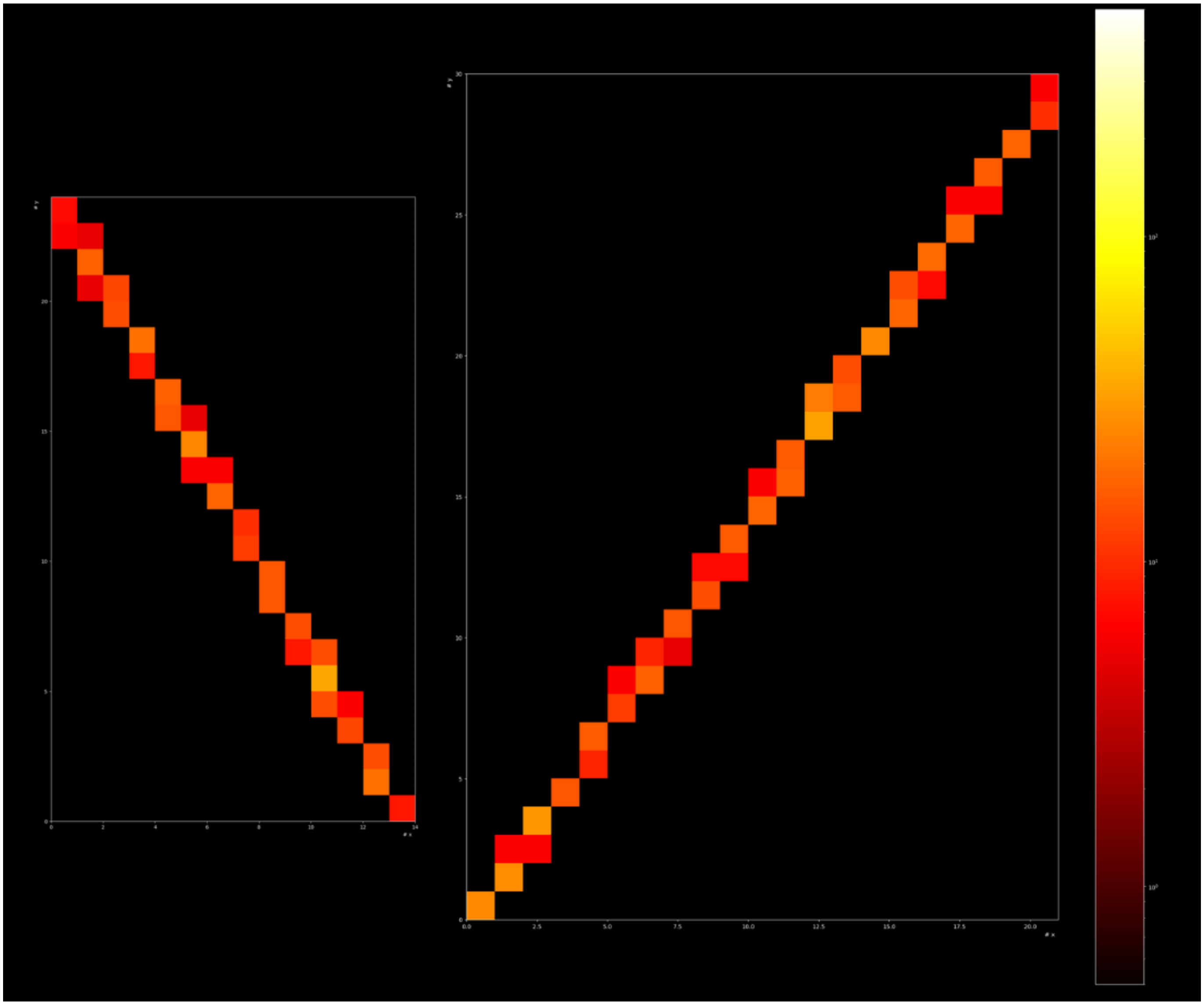


Abbildung 18: Two clear muon tracks

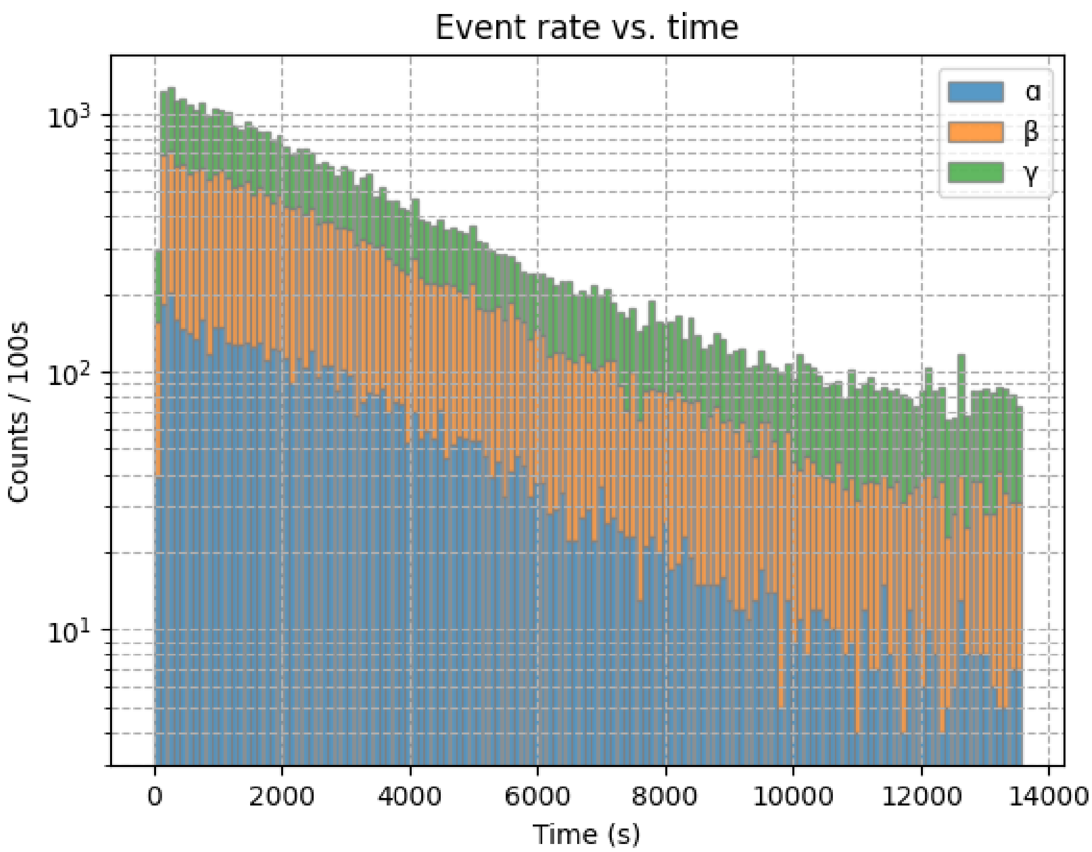


Abbildung 19: Rate of Radon decay products. vs. Time

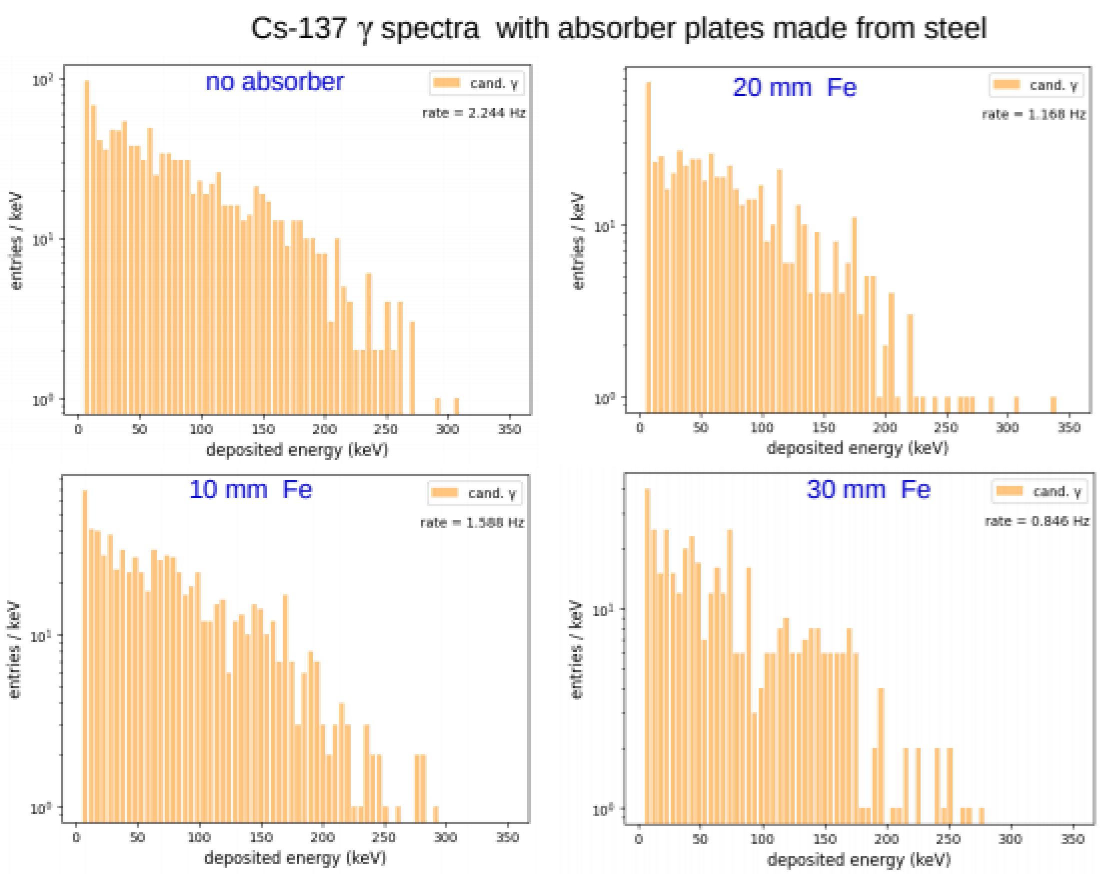


Abbildung 20: Cs-137 γ spectra for different thickness of Fe absorber plates

This behavior indicates a qualitative difference between α and γ radiation. γ rays only interact rarely in matter, and typically only one interaction is seen in thin layers of absorber. The interacting γ transfers its energy to an electron, which then gets absorbed in the material. As a consequence, the γ is removed from the incident beam. As the radiation penetrates a depth l of material, the number of remaining photons $N(l)$ decreases by dN, while the interaction rate is proportional to $N(l)$. This leads to an expected exponential dependence of the remaining number of photons, $N(l) = N_0 \cdot \exp(-\mu\, l)$. μ is the mass absorption (or attenuation) coefficient of the material, which can be determined in a sequence of measurements with varying absorber thickness and incident γ energy. Often, the absorption length, $x_a = 1/\mu$ is used instead of the attenuation coefficient. This exponential behavior is confirmed in the plot below, which shows the measured γ rate as a function of the penetrated thickness of iron absorbers.

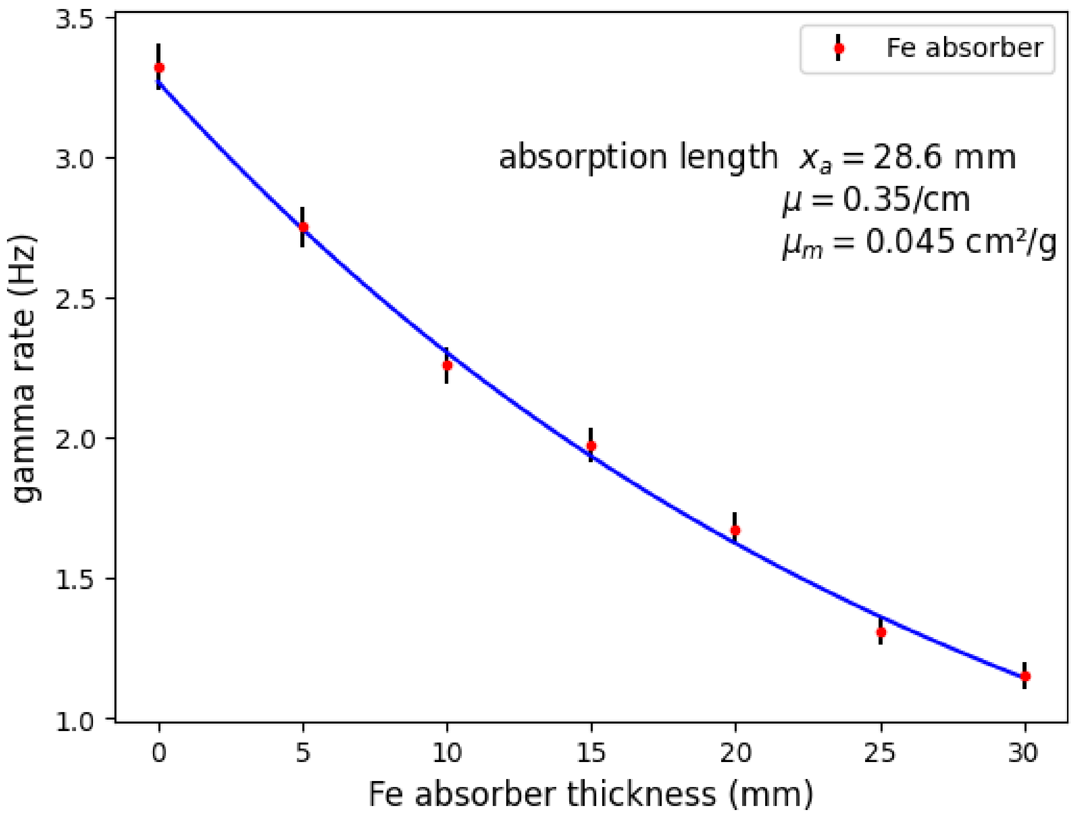


Abbildung 21: Cs-137 γ rates vs. thickness of Fe absorber plates

Note that there was an absorber plate suppressing any α and β particles emitted from the radioactive source. The measurements with additional absorber plates behind the shield confirm the expected exponential relationship between the γ rate and the absorber thickness.

γ-absorption to classify materials Once the special behavior of γ ray absorption is established experimentally, a variation of the experiment demonstrates how to identify materials by their typical attenuation coefficient.
A material sample of thickness d is placed between a γ-source with an Aluminium absorber (to get rid of α and β radiation) and a miniPIX detector. The rates I and I_0 observed with and without the sample directly yield the (linear) attenuation coefficient of the material:

$$\mu = ln(\frac{I_0}{I})/d.$$

If also the density of the probe is known, the mass absorption coefficient, $\mu_m = \mu/\rho$ (in units of cm² / g) can be compared to tabulated values to determine the material type.

Note that the attenuation coefficients determined with the simple set-up proposed here are smaller than the tabulated ones, because Compton-scatted photons are also registered, while - according to a strict definition - they should be excluded. This can be achieved by using collimation tubes allowing only direct paths of γ-radiation from the source to the *miniPIX*.

Sample analysis in *Jupyter* notebook

The *mPIXdaq* package contains a *Jupyter* notebook and some *Python* modules in the subdirectory `analysis/` which illustrates many of the analysis steps described above. Small data samples in the subdirectory `data/` serve as examples to try out and further develop analysis code.

- ***Jupyter* notebook** to analyze cluster data in subdirectory `analysis/`:
  - `analyze_mPIXclusters.ipynb`
  - `LandauFit.py` for fitting a Landau distribution
  - `peakFitter.py` to search for and fit a Gaussian to peaks in spectra
- **Data sets** as examples for the *Jupyter* notebook analysis in subdirectory `data/`:
  - BlackForestStone.yml.gz # *data recorded with a low-activity stone (Uranium ore) from the Black Forest*
  - BlackForestStone.csv.gz # *properties of clusters from the Black-Forest stone*
  - BlackForestStone_clusters.yml.gz # *cluster properties and pixel data from Black-Forest stone*
  - gammaRadiation_clusters.yml.gz # *Black-Forest stone shielded with 3mm Aluminium*
  - ambientRadiation_clusters.yml.gz # *long run with ambient radiation*
  - Radon_clusters.yml.gz # *cluster and pixel data from Radon decay products*

The *Jupyter* notebook contains extensive documentation and should be self-sufficient to understand the analysis methods and algorithms and to adapt them for own studies. After installation of the *Jupyter* environment, the notebook is started fom the command line via `> jupyter-lab analyze_mPIXclusters.ipynb`.

The notebook covers the following topics:

1. Data import to *pandas* data frame
2. Overview of features and data quality checks
3. Classification of clusters
4. Study of α particles
5. Study of β radiation
6. Study of γ interactions
7. Identification of tracks from cosmic muons
8. Rates and energies of α, β, γ radiation
9. Access to and detailed analysis of pixel data
10. Landau distribution of pixel energies along β tracks